Anatomy of Language Form in a Wavelet-like State-Space KAN

A mechanistic case study: every mechanism read from the model's own functions, and a methodological audit of how such claims should be tested

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Abstract

In a byte-level language model whose edges are named time functions, two language-level facts become measurable. First, a boundary clock: at word boundaries the model's input-layer time step grows 2–2.6×, erasing the current word's state, and the trained solution causally depends on it (word-initial damage +1.12–4.02 nats against +0.00–0.35 for position-matched controls across five tiers, n=5 seeds). The clock is byte-keyed, and we report the null that bounds it: on separator-free text the deeper layer still elevates its step at implicit word boundaries in all five seeds (1.21 ± 0.04), but clamping that elevation separates from controls only weakly and inconsistently across seeds — the linguistic-looking signal at depth is observational, not causal. Across architectures the same battery gives the falsifiable contrast: trained Mamba-2 develops boundary gating with the *opposite* operation (it skips boundaries where WSKAN resets), and the transformer's asymmetry inverts entirely — boundary mechanisms are an SSM-family possibility, realized in at least two opposed forms, not a universal one. Second, above the bigram level, word-level content is present in the write path but almost unreadable by the model's own linear readout (held-out $R^2 \lesssim 0.03$) — a scale-conditional observation, not an architecture verdict: matched-size Mamba-2 fails the same recall probe.

Both findings rest on an unusually exact evidentiary layer, and that layer is the paper's second contribution. The substrate — the *Wavelet-like State-Space KAN* (WSKAN) — is built so that mechanism-level claims can be measured rather than inferred: edges are damped-oscillator SSM impulse responses with closed-form kernels, gates are named byte features, and additive residuals with a tied linear head make every decision decompose exactly. On it we audit the findings themselves: the organization is load-bearing for the found solution but not required by the task at any measured training budget, a shuffled-prior control shows that even a *wrong* assignment of the same structure retrains for free, and an architecture-level scramble of the feature basis leaves the clock intact — within this parameterization family, the organization is a property the parameterization offers, not one the data demands; the cross-family edge is suggestive only. Scope: small-scale byte-level models, one KAN family, three datasets, up to ≈10m parameters. Code and checkpoints: `https://github.com/llaa33219/wskan`; released weights (15 models, safetensors, real generation outputs on each card): `https://huggingface.co/llaa33219`.

1 Introduction

Interpretability research on sequence models faces a measurement problem: the objects we inspect (attention heads, neurons, probes) are not the objects the model computes with, and post-hoc

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attributions are known to be unreliable. This paper takes the opposite route — build the model so that its computational units *are* named functions, then read them. The claim is deliberately narrow: at small byte scale (1k–10m parameters), on one architecture family, a complete mechanistic account is possible end-to-end; what is offered as transferable is the audit protocol, not the anatomy. The strongest current mechanistic results sit at 370m–2.7B parameters; nothing here transfers across that gap by default.

A note on the comparison class: WSKAN is a *Kolmogorov–Arnold language model* — learnable functions on edges, addition-only nodes — not a transformer-family model. On the KAN-LM axis (spline edges, wavelet edges, and the B-spline edge audit of [11]) this is, to our knowledge, the first member with *temporal, recurrent* edge functions and a complete edge-level anatomy; its scale is the scale at which KAN LMs are currently trainable, not a shrunken transformer. We offer no theory for why damped oscillators should be a privileged edge basis for language — the substrate is an instrument choice, and every claim is conditional on the family; §7 measures how much that conditionality matters. The contribution is not the model's language skill and not a new architecture to adopt: it is *a readable substrate plus the audit protocol that tests claims on it* — the substrate (edges as named time functions, gates as named byte features, additive residuals with a tied linear head) and the battery that makes claims checkable (exact decision decomposition, causal interventions, surgery-vs-retrain separation, null baselines). Concretely measurable here and not in a transformer or stock SSM: what did each edge learn (a closed-form kernel per edge), which bytes cause a decision (exact per-term decomposition), how wrong is a standard post-hoc attribution (measured against that ground truth).

Reader's guide. The paper has a spine: **build** the substrate (§2), **read** it (§3–§6), **audit** the reading (§7–§8).

§5–§6 are the two boundary results (what is not represented; what can be derived by hand). §7 audits where the organization comes from — the section that changed our own claims the most. §8 scores post-hoc methods against the exact ground truth. Appendices carry the tables; the main text keeps only the numbers that carry the argument.

WSKAN (Wavelet-like State-Space KAN) parameterizes each Kolmogorov–Arnold edge as the impulse response of a stable complex state-space model (SSM). One parameterization gives two equivalent views: a static wavelet per edge (closed form) and a recurrent selective scan (Mamba-style). Because the edge is a damped oscillator with named parameters — decay σ, frequency ω, a per-mode timescale ladder ρ_k, an input-driven dilation field Δt — the trained model can be dissected with exact arithmetic rather than probes alone.

What the paper claims (three claims; the rest is support). If the organization is barely required, why dissect it? Because this is the only artifact we know where what gradient descent finds easy can be read off exactly. The anatomy below is that reading.

(i) The boundary clock. The clock's form is offered by the design (input-driven step, byte-feature gates); what is learned is the routing — boundary bytes get large steps — and what we claim is the *dependence*: an input-driven clock causally gates word memory (word-initial damage +1.12–4.02 nats against +0.00–0.35 for position-matched controls across the five tiers, $n = 5$ seeds). It is byte-keyed: the deeper layer's elevation at implicit boundaries replicates across seeds but is causally inert under clamping (§3.2).

term	meaning
clock / $\Delta t_{n,i}$	input-driven per-position step size (dilation field)
warped time	its accumulation $T_{n,i} = \sum_{j \le n} \Delta t_{j,i}$
mode k	one complex oscillator (decay σ, frequency ω) of an edge
ladder ρ_k	learned per-mode timescale multiplier
gate split	write/read gates factorized into named byte features + content
edge function	the per-edge kernel $\psi_{io}(t)$ — the KAN weight, as a function
substrate	the model, used as a measurement platform
gauge symmetry	parameter re-indexing that changes nothing observable
surgery vs retrain	intervention on the trained solution vs training without the structure

Table 1: Glossary of terms used throughout.

(ii) Content barely reaches the linear readout (scale-conditional). Above the bigram level, word-level content is present in the write path but exposed only weakly to the model's own linear readout (held-out $R^2 \lesssim 0.03$ at every tier; §5) — a statement about these scales, not about the architecture: matched-size Mamba-2 fails the same recall probe.

(iii) The audit. The organization is load-bearing for the found solution but not required by the task at any measured training budget, and a shuffled-prior control shows a wrong assignment of the same structure is equally free (§7) — within this family, the organization is offered by the parameterization, not demanded by the data. Table 1 fixes terminology; Table 2 lists the claims with their evidence, scope, and falsifiers.

Why this architecture enables the measurements. Three design choices do the work. (a) *Edges are named time functions*: each edge is a small bank of damped oscillators — six complex modes plus gains, a dozen numbers — with closed-form kernels, so "what did this edge learn" has a concrete answer — a set of decays, frequencies, and gains — rather than a row of a matrix. (b) *Feature-factorized gates*: the structural part of every write/read gate is a lookup over eight named byte features, so gate behavior is read, not probed. (c) *Additive residuals with a tied linear head*: the final logits are a sum of named path contributions, so decisions decompose without approximation. None of these is available simultaneously in a transformer or a stock SSM, where the equivalent objects are entangled in unstructured projections.

What the paper does not claim. Performance: Mamba-2 leads in 12 of 15 benchmark cells; of the rest, WSKAN leads 1m UltraChat clearly and 1m WikiText by 0.0002 nats (a tie within seed noise), and the transformer leads 1m TinyStories and 10m WikiText (Appendix C). (§7 covers why the model is still worth reading — the architecture is not proposed for adoption, and nothing in the paper argues otherwise). The exchange, quantified: at the 100k and 10m tiers +0.02–0.12 nats of CE against matched baselines (more at the smallest tiers: +0.15–0.29 at 1k–10k) buys an artifact on which decisions decompose exactly, interventions have named targets, and a hand-written rebuild runs at all (§6).

2 Model

The model family is the eleventh iteration of the project.[1]

[1] WSKAN ≈ `wskan11` in the scripts; the `V1…W12` prefixes mark era, not architecture branches.

claim	evidence	scope	falsified by
byte-keyed boundary clock gates word memory	causal battery, 5 seeds, +1.12–4.02 vs +0.00–0.35 nats vs control	this family, ≤10m	removal not hurting word-initial prediction
decisions decompose exactly	integrity assert $< 10^{-4}$ on all decomposed contexts	any additive-residual + linear-head model	a failed sum
content barely reaches the linear readout above bigrams	linear probes, per-tier (§5)	tiers measured; readout structure	reference-level variable readable by a linear probe
organization load-bearing, barely task-required	surgery vs retrain, per-tier (§7)	this parameterization family	a structure-free retrain that fails

Table 2: Claims at a glance.

2.1 Edges as stable oscillators

Each edge $(i \to o)$ carries N=6 complex modes with $\lambda_{ik} = -\sigma_{ik} + i\,\omega_{ik}$, rescaled by a learned per-mode ladder ρ_k: $\tilde{\lambda}_{ik} = \rho_k \lambda_{ik}$. Stability is by construction ($\sigma > 0$). At initialization $\sigma \equiv 0.5$, $\rho \equiv 1$ (flat), and $\omega = \pi \cdot [1, \ldots, 6]$ (a harmonic skeleton that retraining experiments show is unnecessary; §7).

On the name. The edge kernels are damped-oscillator impulse responses — wavelet-*like*, not strictly admissible wavelets, and the family name now says so (WSKAN = Wavelet-like State-Space KAN; the acronym and all artifact names are unchanged). The strict zero-mean condition is not enforced and is only approximately satisfied in the trained model (|DC|/RMS per edge over the first 5 warped-time units: median 0.14; 34% of edges below 0.1 — and the ratio falls with horizon, median 0.007 over 2,000 units). Nothing in the paper depends on the strict condition.

2.2 The content-warped scan

With input-driven step $\Delta t^{\mathrm{dyn}}_{n,i} = \mathrm{softplus}(W_{dt}\,\mathrm{norm}(x_n))_i$ and ZOH-consistent write, the per-channel recurrence

$$h_{n,ik} = e^{\tilde{\lambda}_{ik}\Delta t^{\mathrm{dyn}}_{n,i}}\, h_{n-1,ik} + B_{n,ik}\,\Delta t^{\mathrm{dyn}}_{n,i}\, x_{n,i}$$

(the exact ZOH discretization would carry the factor $(e^{\tilde{\lambda}\Delta t}-1)/\tilde{\lambda}$; we use its first-order form $B\,\Delta t\,x$ and let the learned B absorb the remainder. Stability cannot be affected by the write's form: the multiplier obeys $|e^{\tilde{\lambda}\Delta t}| = e^{-\rho\sigma\Delta t} < 1$ for any $\sigma > 0$, $\Delta t \geq 0$, so the state is contractive regardless of what is written into it. The scan computes exactly the recurrence above — no approximation — and the closed form below is the exact solution of that recurrence, not of the continuous-time system) has the exact closed form, with warped time $T_{n,i} = \sum_{j \leq n} \Delta t^{\mathrm{dyn}}_{j,i}$:

$$h_{n,ik} = \sum_{m \leq n} e^{\tilde{\lambda}_{ik}(T_{n,i} - T_{m,i})}\, B_{m,ik}\,\Delta t^{\mathrm{dyn}}_{m,i}\, x_{m,i}.$$

Token distance is not position distance; it is accumulated dilation — this is the one place where the model departs from a standard selective scan, and the source of everything timed in §3.2.

The write/read gates are feature-factorized,

$$B_{n,ik} = \sum_f \alpha_f(\text{byte}_n)\, M^B_{f,ik} + W^B_{\text{lr}}\, x_n,$$

where B is the per-position write gate (computed) and M^B the named table (a parameter), with fixed named byte features $\alpha \in \{\text{space, newline, punct, upper, lower, digit, vowel, 1}\}$ — the last entry is a constant bias feature — plus a low-rank content residual. The read gate $C_{n,ik}$ has the identical form with its own table M^C and residual. The mode ladder is $\rho_k = e^{[\log \rho]_k} > 0$ by construction, so $\tilde{\lambda} = \rho_k \lambda$ keeps negative real part and stability is unconditional. The layer output is $y_o = \Big(\sum_{i,k} C_{ik} \left(\text{Re}[h_{ik}]\, a_{iok} - \text{Im}[h_{ik}]\, b_{iok} \right) \Big) \odot \text{SiLU}(W_z x) + x W_{\text{base}}$, where the readout gain of mode (i,k) into output channel o is the complex $g_{iok} = a_{iok} + i\, b_{iok}$: per (input channel i, mode k), a complex state h_{ik} is read out through the per-output gains and summed. The canonical model is a 2-layer residual stack with a tied byte-embedding head (depth and width vary by tier — 1–6 layers, d 4–512; Appendix E), and training uses a fused associative-scan kernel (verified against a serial loop to $\leq 4{\times}10^{-6}$). All dynamical quantities are reported in warped-time units; per-mode tuples satisfy the identities $Q = \tilde{\omega}/(2\tilde{\sigma})$, $t_{1/2} = \ln 2/(\tilde{\sigma}\, \bar{\Delta} t)$ exactly (with $\tilde{\sigma} = \rho\sigma$, $\tilde{\omega} = \rho\omega$ the effective per-mode parameters), while medians of these correlated quantities do not (one of the corrections listed in §7).

3 The anatomy

3.1 The byte manifold organizes by linguistic function

The learned 256-way byte embedding is functional: within-class cosine 0.47 vs between-class 0.07; nearest neighbors are vowels, case variants, separators, adjacent digits.

3.2 The boundary clock

Terminology, once: the *clock* is the per-position dilation field $\Delta t_{n,i}$; *warped time* $T_{n,i} = \sum_{j \leq n} \Delta t_{j,i}$ is its accumulation ("accumulated dilation" says the same thing operationally). Two ratios recur below: the *per-class* Δt ratio (boundary bytes vs lowercase letters, canonical checkpoint) and the *clock ratio* of the all-sizes battery (mean Δt at spaces vs letters, per tier, 5 seeds).

Measured on the canonical checkpoint: word-boundary bytes advance the clock 2–2.6× faster than letters in every layer (space $\Delta t \approx 0.21$–0.26 vs lowercase ≈ 0.10; the battery's 5-seed metric reads 1.95 ± 0.27 at this tier — the same field under a different normalization; the claim rests on the ratio's presence and seed-stability, not its exact value), so a boundary erases exponentially more of the current word's memory. The causal battery (5 seeds; controls: letter positions perturbed by the same magnitude, position-matched) shows this is load-bearing: removing boundary Δt pulses degrades word-initial predictions by $+2.87, +2.29, +1.43, +1.89, +4.02$ nats across the five seeds, against $+0.13, +0.13, +0.12, +0.12, +0.15$ for the matched letter controls (each seed: 16,384 eval positions, ≈2,600 word-initial decisions; the letter control is count-matched to the boundary positions, so the tight control spread is precision from that base, not a small sample), and injecting false pulses creates word-ending behavior (sufficiency). Two strengthenings: the necessity clamp holds on all five seeds (L0 word-initial damage $+2.57 \pm 0.48$ nats), and the clock is not merely byte-keyed — on space-free text (separators removed, true boundaries recorded from the source text, so no label circularity; the model, trained with separators, is out-of-distribution on this input), Δt still elevates at implicit boundaries at L1 in all five seeds (ratio 1.21 ± 0.04; L0 stays at noise, 1.05 ± 0.03).

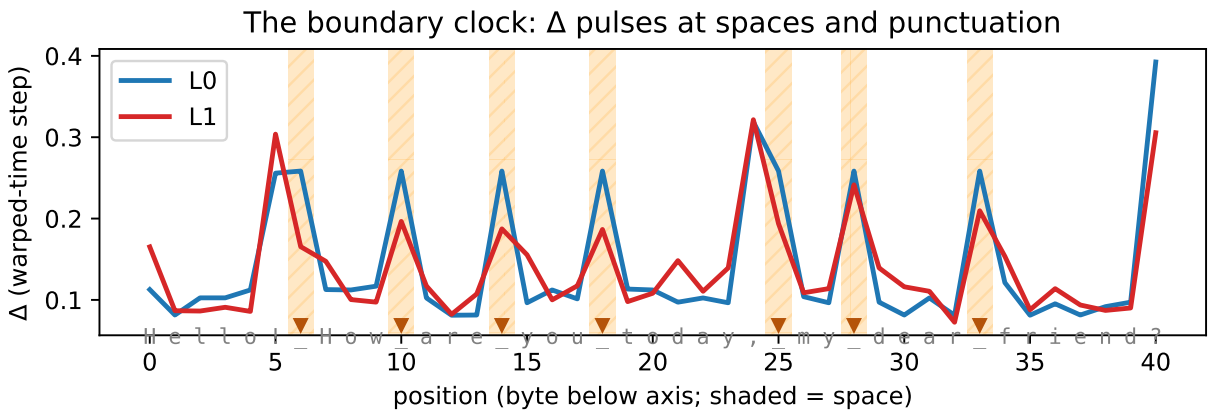


Figure 1: The boundary clock on a held-out sentence: Δt (mean over channels, both layers) pulses at spaces and punctuation (hatched bands and triangle markers = space positions). *The clock is real and byte-keyed — the model ticks faster at word boundaries.*

The causal version of the implicit-boundary test is weaker: clamping Δt at implicit boundaries damages word-initial prediction more than position-matched controls on average ($+0.24 \pm 0.14$ vs $+0.15 \pm 0.15$ at L1) but within seed variance — three of five seeds separate cleanly, two do not. The clock is byte-keyed at the input layer and carries a seed-robust *observational* linguistic signal at depth with weak, seed-inconsistent causal support — measured, in both directions.

Cross-architecture. Trained Mamba-2 develops boundary-responsive gating too — with the *opposite operation*: its per-head Δt at spaces drops to 0.13–0.65× the letter mean, and its ZOH write magnitude to 0.15–0.70× in the deep layers (the input layer goes the other way, 1.4×). Both models are parameter-matched to WSKAN (tier configurations in Appendix E), byte-level, same data, same training budget; the mamba2 granularities differ by design (per-head pooled Δt vs per-channel). Mamba-2 *skips* boundaries; WSKAN *resets* at them. The comparison is causal on both sides: forcing Mamba-2's Δt at spaces up to the letter mean damages word-initial prediction +0.28 nats vs +0.08 at letter positions (a matched control intervention shows the reverse asymmetry, +0.19 vs +0.29) — position-specific, though less sharply localized than WSKAN's clock. And the attention baseline inverts the asymmetry entirely: masking the transformer's attention to boundary bytes costs word-initial CE +2.93, while masking letters costs +4.75 (0.62×) — no special boundary dependence. The comparison is not an artifact of WSKAN being underfit: at matched eval CE (Mamba-2 at 75k steps, 1.182 vs WSKAN's 1.184), the same battery gives +0.24 word-initial damage for boundary removal against +0.18 for the matched control — the position asymmetry survives loss-matching, so the sharpness gap between the two architectures' boundary mechanisms is architectural, not a maturity artifact. Boundary-responsive gating therefore appears in both SSM members we tested — with opposite operations and different Δt granularity (per-head pooled vs per-channel). With two members and a structural granularity confound, this is an existence observation, not a family-level conclusion: the SSM family *admits* boundary mechanisms, in at least two opposed forms; whether language demands one is not answered here — a correction to our earlier framing, measured by the same battery on all three. The causal protocol is position-matched on both SSM architectures (raise Δt at the same positions, measure the same quantity); what differs is the architecture's own aggregation granularity (Mamba-2 pools 16 channels per head, WSKAN does not pool) — that difference is part of what is being compared, and we flag it as an

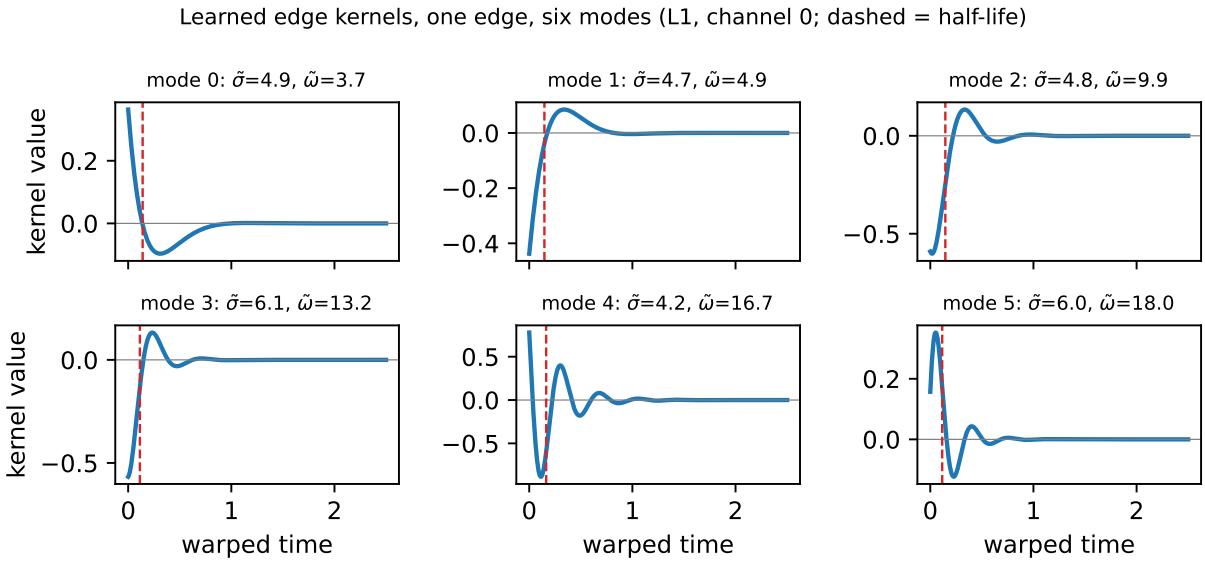


Figure 2: One edge's six learned mode kernels (L1, channel 0, canonical checkpoint) in warped time; dashed lines mark each mode's half-life. Frequencies rise across modes; decays vary less. *The oscillator bank is a frequency ladder, learned.* Population statistics exist elsewhere (|DC|/RMS in §2; ladder R^2 in Appendix B); this gallery shows what a single edge looks like.

interpretation of the sharpness gap, not a measurement.

3.3 The oscillator bank

The learned effective timescales $\rho_k \sigma_{ik}$ span a ladder (geometric-fit R^2 0.64–0.89 on the canonical tier's layers, from a flat init; the full per-tier grid gives 0.60–0.97, Appendix B; ratio 1.3–2.3×) whose slow tail reaches ∼3,190 tokens at the 10m tier while the median half-life stays ≈ 1 token at 100k. The tail carries $< 1\%$ of readout weight — present, unused.

3.4 Gates split structure from content

The named rows of M^B, M^C carry the discrete formatting bytes (digit, newline, punctuation); the residual carries byte identity (R^2 1.00 → 0.76 → 0.50 with depth) and bigram statistics (R^2 0.91/0.84). At 10m the write path's non-bigram share is predominantly *trigram-level* statistics (cumulative explained variance 0.48 → 0.65 → 0.94 over byte/bigram/trigram families; word identity adds 0.09; 5.4% unidentified) — scale growth is the *order* of local statistics, not a new object class.

3.5 Decisions decompose exactly

Additive residuals plus a linear tied head imply every decision margin decomposes into named summands whose sum equals the margin *exactly* — here and throughout, "exact" means no approximation in the decomposition arithmetic: each summand is a computable part of the forward pass, and the parts sum to the margin within floating-point tolerance (asserted $< 10^{-4}$ on every use).

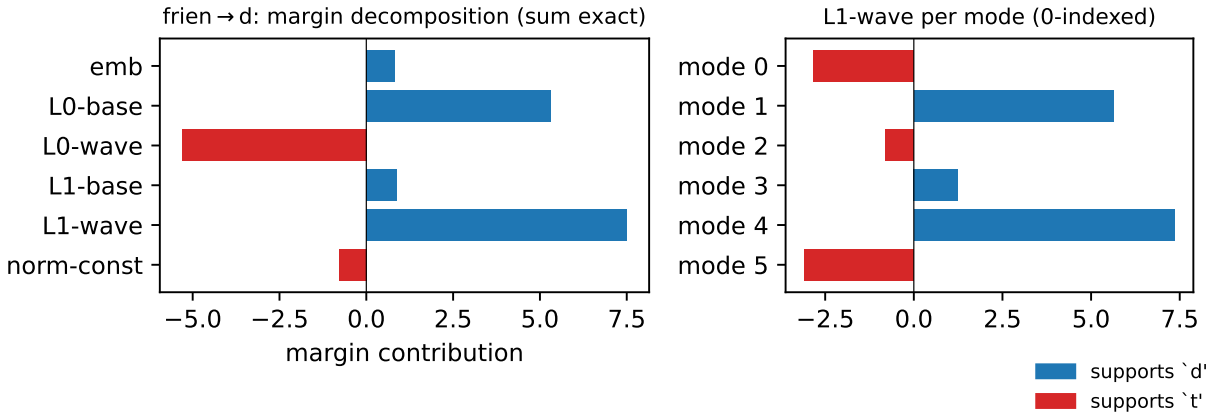


Figure 3: The same decision visually. Left: the exact margin decomposition (parts sum to 8.44). Right: the L1-wave term per mode — modes 4 and 1 (0-indexed) carry the completion here; mode identity is per-decision, not a global semantics. *Decisions are readouts of named parts, not black-box logits.*

Why the sum is exact (and what it does not claim). Let $x_{\text{fin}} = \text{emb} + \sum_l (\text{base}_l + \text{wave}_l)$ be the final residual stream. Every summand in the table is a vector the forward pass itself computed — the layer output out_l is *defined* as $\text{base}_l + \text{wave}_l$, and the final LayerNorm is affine given the batch statistics: $\text{norm}(x) = \sum_v (v/\sigma) \cdot g + (\text{const})$ for any additive decomposition $x = \sum_v v$. The table is therefore a distribution of a sum the model actually performed — it is exact for every decision, not because any cancellation holds, and the integrity assert ($< 10^{-4}$) is checked on every decomposition we report (the frien→'d' case, the 50-per-tier truncation contexts, and all 150 benchmark contexts). What it does *not* claim is causal independence: wave_1's value already contains L0's influence through the layer input, so a summand is the model's own computed part, not a counterfactual contribution — causal claims live in the intervention battery (§3.2), which measures removals, not summands.

Canonical case `frien→`d'` (margin 8.44 over the runner-up 't'). The decomposition's novelty is not the additivity — any additive-residual model has that — but that here every summand is a *named* object (a mode's kernel, a gate's feature row), so the parts are readable, not just computable:

summand	margin contribution
token embedding	+0.83
L0 base (skip)	+5.32
L0 wave (scan)	−5.31
L1 base (skip)	+0.88
L1 wave (scan)	**+7.50**
norm constant	−0.79
sum	**+8.44** (exact)

The decisive L1-wave term further decomposes per mode as [−2.85, +5.64, −0.81, +1.25, +7.36, −3.09].[2] Mode identity is per-decision, not a global semantics: across 20 pre-registered completion

[2]Rounding and precision notes. At two decimals the parts sum to 8.43; the three-decimal parts +0.829, +5.321,

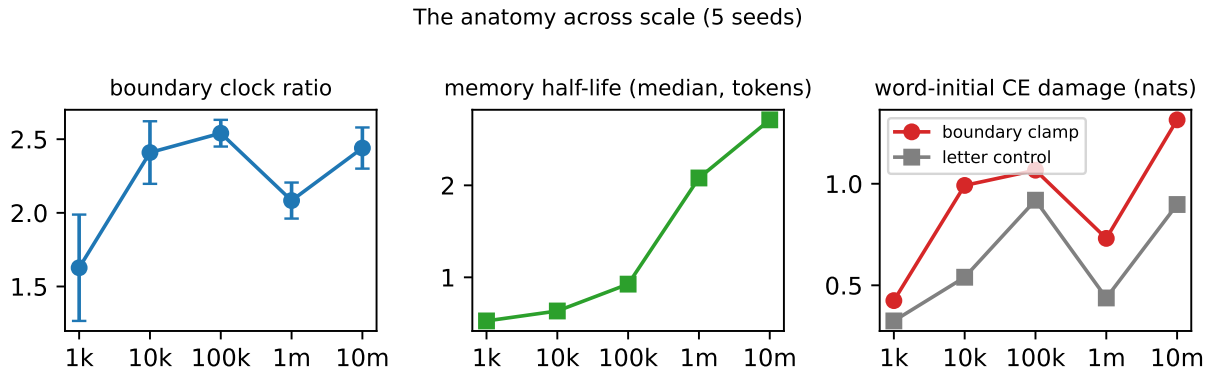


Figure 4: The anatomy across scale (5 seeds): the clock ratio stays 1.6–2.5, memory grows with capacity, and the causal gap over the matched control holds at every tier. *The anatomy is not a single-scale story.*

contexts, the dominant path and modes vary (§6).[3] The recurring pattern across decisions: word endings resolve as L1-over-L0 disagreement resolutions (the shallow wave path resists, the deep one completes). The same decision across tiers: 1k fails outright (predicts a space, margin 0.42), 10k succeeds by a hair (margin 0.08), then the margin strengthens 8.44 / 12.86 / 13.80 at 100k / 1m / 10m — morphological completion is the same mechanism, sharpened by scale.

The audit across tiers. The organization audit of §7 is not 100k-only: the surgical battery (all tiers) and the retrain test (10k, 1m, 10m, seed 42) show the same verdict at every scale — the named tables are surgically load-bearing at small scale and nearly free at 10m (+2.5 → +0.05 nats), the gauge symmetry is exact everywhere, and retraining without the structure stays within 0.031 nats of the baseline at every tier tested. The detectability bounds are small in absolute terms (0.04–0.10 nats, roughly 3–8% of CE at these tiers) — null results are bounded, not exact zeros.

3.6 Production

In generation the same loop runs from the inside: the clock pulses at boundaries, word-internal letters are wavelet-path decisions against a negative token prior, and punctuation triggers a synchronized spike across clock, paths, and modes.

4 How the anatomy varies with scale (5 tiers × 5 seeds)

The anatomy is consistent across the tiers measured (within this KAN family, up to 10m — nothing is claimed beyond it; the full 5-tier table is Appendix B). The surgical/retrain audit (§7) runs at every tier for the surgery and at 10k/1m/10m for the retrains; only the circuit-level detail is 100k-only.

tier	Δt ratio, space/letter (L1, 5 seeds)	half-life med (tok)	`frien`→`` `d' ``
1k	1.63 ± 0.40	0.53	0/5
100k	1.95 ± 0.27	0.92	5/5
10m	2.48 ± 0.14	2.45	5/5

The causal word-initial effect exceeds its matched control at every tier: +1.12–4.02 nats of damage against +0.00–0.35 for the control ($+2.50 \pm 1.00$ vs $+0.13 \pm 0.01$ nats at 100k). The differences, not any ratio, are the headline: the near-zero control damage makes ratios denominator-sensitive. (The Mamba-2 comparison of §3.2 — position-specific on both, sharply localized only in WSKAN — is not repeated here.) The factor dips at 1m — and the dip is causal, not metric: clamping the boundary Δt to the letter mean at *each single layer* damages word-initial CE on *every* layer of the 1m tier (+0.13–0.39 across L0–L5, s42, largest at the first and last layer), while at 100k the same clamp is concentrated in the input layer (+3.19 at L0 vs +0.80 at L1). The clock's causal role redistributes with depth: distributed across all six layers at 1m, concentrated at the input layer at 100k. The tier grid confounds depth with capacity (the 1m tier is 6 layers, the 10m tier 2), so we ran the control: a 6-layer model at 100k-tier capacity (109k params, 2 seeds) shows the same clock at every layer (Δt ratios 1.4–3.2) with the causal role partially spread but still input-layer-heavy (L0's share of the total clamp damage 29–43%, against 80% for the 2-layer 100k and 19% for the 6-layer 1m) — depth spreads the role, capacity dilutes it further, and the input layer remains the largest contributor at matched capacity (`W12_depth6_analysis.py`). The second crossing of the grid agrees: a 2-layer model at 1m-tier capacity (960k params, 2 seeds) sits between the two, L0 share ≈60% — depth is the dominant axis of the redistribution, capacity a secondary dilution (`W12_variant_clocks.py`). The clock itself is present at every layer of the tier (per-layer space/letter Δt ratios 2.18 / 1.86 / 1.52 / 1.56 / 1.61 / 1.75). The memory tail grows $4 \to 3{,}190$ tokens; morphological competence appears only $\geq$ 100k. The wavelet's performance contribution — measured against WSKAN-real, the identical architecture with the oscillation frozen off ($\omega \equiv 0$); 73 of 75 paired cells — is real, modest, and scale-shrinking: $-0.02 \ldots -0.08$ nats at 1k–100k, $-0.01 \ldots -0.03$ at 1m–10m.

−5.306, +0.879, +7.504, −0.789 sum to 8.438 against a margin of 8.4392, and the per-mode list sums to 7.504, matching the L1-wave part exactly. All measured in fp32; the same forward pass in the bf16 training precision gives a margin of 8.30, a 0.14 precision gap — every analysis number in the paper is fp32. The L0-base/L0-wave near-cancellation is a coincidence of this decision: L0's residual output is 2.6× its input norm here, far from identity; the opposing votes are strong and happen to balance — they are not a conserved quantity.

[3] A note on naming: "modes 4 and 1 carry the morphological completion" (0-indexed) is a statement about *this* decision — the context ends in `frien` and the answer completes `friend`, so the positive modes are the ones driving the completion here. The decomposition is exact arithmetic; the semantic label attaches to the case, not to the mode — the named modes here illustrate the machinery at work on a real decision; they are not claimed as a mode semantics.

5 What the model does not represent linearly

Definition (operational, scale-conditional). *Form* = the distributional regularities of the text surface — boundary timing, local n-gram statistics, morphological continuation. *Content* = reference: facts, entities, long-range meaning. The boundary is our choice, marked rather than hidden; it is conditional on the tiers measured (up to 10m, where content has loosened to trigram order, no further).

Morphological completion sits exactly on the boundary. We count it as form because its measured machinery is local — a boundary clock and a per-decision mode vote over a handful of recent bytes — with no factual content; we do not claim morphology is *only* surface structure, and a reader who places it on the content side loses none of the measurements, which are definition-independent. The boundary is also falsifiable: if a future tier shows reference-level variables (facts, entities, turn state) in the linearly accessible write path, the "form-only" claim dies; the 10m measurement (content loosens to trigram order, no further) is where it currently stands. What the battery does *not* cover: facts, entities, coreference, and other long-range semantics — untested at every tier, and the honest reading of the 10m trigram finding is that reference has not yet entered the local window, not that it cannot at larger scale or on other tasks. The label names the measured phenomenon compactly — it does no work beyond that, and deleting it changes no number below. Sensitivity: if morphology is moved to the content side, the claims that move are the label "form-only" and the morphological-completion examples; the clock, the ladder, the gate split, the decomposition machinery, and the exclusion results below all stand as measured.

Linear probes on the write path, measured at every tier (UltraChat, seed 42; script in Appendix F):

predictor (linear R^2)	1k	10k	100k	1m	10m
byte identity	0.80	0.83	0.44	0.35	0.48
bigrams	0.81	0.76	0.64	0.53	0.64
word identity	0.10	0.13	0.15	0.14	0.14
(in-sample shuffled-label null)	0.08	0.08	0.08	0.08	0.08
(ridge, held-out, best α)	−0.00	0.01	0.02	0.02	0.02
sentence position (8-bin)	0.05	0.03	0.01	0.01	0.02
word identity, MLP held-out	0.07	0.10	0.26	0.29	0.25
(shuffled-label control)	0.04	0.03	0.02	0.01	0.01

Above the bigram level the pattern is scale-stable, and null calibration sharpens rather than rescues it: word identity's in-sample R^2 (0.10–0.15) sits just above its shuffled-label null (0.08 at every tier — with 500 classes on 4k positions the in-sample null is high, a calibration the table now reports), and a held-out ridge sweep puts the linearly decodable share at $R^2 \lesssim 0.03$ at every tier, against held-out R^2 up to 0.60 for byte identity and 0.20 for bigrams. The small value is not probe overfitting: quadrupling the probe positions (4k → 16k) leaves it in the same band (held-out R^2 0.029–0.032 at 100k and 10m, accuracy 3× the shuffled null, while the in-sample null falls 0.08 → 0.04 as expected). Sentence position is absent (≤ 0.05), and the nonlinear recovery grows from toy levels (7%) at 1k to ≈25–29% at 100k–10m and saturates. At 100k we additionally probed the post-gate signal and the residual stream: word-identity linear R^2 stays 0.15–0.16 (the gate does not linearize), while the MLP recovers 23.4%/21.5% there. Nonlinear exploitability by downstream gates remains open at every tier. This linear-fails/nonlinear-succeeds pattern is not specific to this model — it is the expected relationship between probe classes — but here the readout structure is known, so the pattern pins down a sharper statement that generic substrates cannot support:

word-level information is present, but the model's own linear readout can use almost none of it. We scope the statement precisely, because the construction argument is per-layer: the readout C is linear in its own layer's state by construction, but at depth ≥ 2 the end-to-end state→logit map is *not* linear (the SiLU gate and the LayerNorm see earlier layers' outputs). The claim is therefore empirical, not architectural, and it covers the whole path the head reads: the linearly decodable share of word identity is small from the write path, the post-gate signal, and the residual stream alike (in-sample $R^2 \approx 0.10$–0.16 against a shuffled null of 0.08; held-out ridge $R^2 \lesssim 0.03$ — the gate does not linearize). A direct alignment check makes the same point from the readout side: the model's own sensitivity to the write path (the mean loss-gradient direction over 2k positions, 100k tier) sits at $|\cos| = 0.14$ with the probe's principal word-identity direction — about 2× the 240-dimensional null of 0.065 — so the readout touches the word-identity direction weakly rather than not at all. (An earlier draft's absolute wording — "never exposed" — is corrected accordingly: the measured statement is "weakly exposed, an order of magnitude below the bigram level".) Decodable is not exploited: whether downstream nonlinearities make use of it is open; the behavioral recall probe (memorize-then-recall) fails identically on matched-size Mamba-2, so "no facts, no recall" is a scale-level observation, not an architecture-level one. A positive control now backs that reading: on an induction (in-context copying) task the measurement does detect signal (4–5% accuracy against a 0.4% chance level, n=200, on both architectures) — so the probe can see recall-type behavior when it exists, and at this scale even local copying sits near the floor. The recall null is a capability floor of the scale, fully consistent with the 370m IOI result [15], where the same model class does store entities linearly. The stronger wording an earlier draft used ("provably") is retracted with the rest in §7. (The word-initial decisions used in §8 sit on the same boundary: predicting a word's opening letter is distributional, not referential, by the same operational criterion.)

Scope note on scale: the table above is already per-tier; at 10m the linearly accessible content loosens one n-gram order (the write path's non-bigram share is predominantly trigram-level; §3). Trigrams stay *form* by our criterion — higher-order but still local distributional statistics — so the definition does not move at 10m; what moves is its order.

6 Deriving the next token by hand

An itemized from-scratch forward pass (serial scan, explicit warped-time history sums, every gate value printed) reproduces the model's logits to 3.8×10^{-6} at the 1k/10k/100k tiers. At 1k (d=4, one layer) the whole computation fits on a page: 576 history terms for a 24-byte context. Worked example: after ``` ``sunrise.\n O'' ``` the model outputs 'n' — the decisive term is the period–newline pair closing the previous sentence, read through mode 4 at warped distance 0.628 (+5.37 of the wave path's +9.26). A near-miss (`"`Ow! Tha"` → 'n' for "Than" where the text has "That") is equally legible.

The decision is a low-rank event at every scale: ranked by contribution, the median context needs 12 / 26 / 6 / 1 / 2 history terms (of 576 / 1,728 / 11,520 / 17.7M / 37.7M) for the argmax to stabilize at 1k / 10k / 100k / 1m / 10m — the median decision at 10m is settled by *two* of thirty-seven million history terms. Static parts alone decide 10 / 14 / 26 / 50 / 30% of contexts — at 1k history is the decision; at 1m half the decisions are already settled before history sums. What this is *not*: derivation without execution. The gates are input-dependent and must be computed; what exists is *transparent simulation* — complete at 1k/10k, truncated at 100k+ (a median 6 of 11,520 top-ranked terms then suffice to stabilize the argmax, Appendix D).[4]

[4]Two clarifications. First, practical scale: the complete 1k case is 576 scalar terms — a full hand-check is hours of careful arithmetic, but every step is printed and checkable; the median case needs 12. Second, the clock is not

A pre-registered dictionary test at 100k (4-byte suffixes with ≥ 8 observations and $\geq 75\%$ next-byte purity, 20 eval contexts): the model matches the corpus completion 20/20 — but the dominant path varies (L1-wave only 8/20) and the canonical mode dictionary transfers to just 2/20. The object-level anatomy transfers; the per-decision semantic inventory does not.

6.1 Writing the model by hand

The strongest test of an anatomy is constructive: can the edge functions be *written* by hand? We hand-set every parameter of a 1k-tier shell from the measured anatomy alone — the clock ratios, the ladder, the gate split, a word-length integrator mode, a sentence memory — with no training (script `W12_handcraft.py`; details in the handcraft report). The result produces pseudo-text with real word boundaries, a plausible length distribution, vowel/consonant texture, and case and punctuation rhythm. The word-length mechanism is visible in a single number series: the margin $\text{logit}(\texttt{space}) - \text{logit}(\texttt{e})$ after k consecutive letters runs $-11.44, -8.26, -6.15, -3.81, -1.41, +2.91$ for $k = 1, 3, 4, 5, 6, 8$ — boundary pressure building as the word lengthens, the clock-plus-integrator pattern of §3.2 working in a model that has never seen data.

configuration	CE (TinyStories eval)
hand-written, static skeleton only (emb + base + frequency-tilted head)	7.53
hand-written, full (static + form machinery)	**5.45**
unigram reference	3.06
trained 1k model	2.23

The decomposition of the score is honest in both directions: the form machinery is worth -2.08 nats over its own static skeleton, and the skeleton itself is a bad prior (7.53 against the 3.06 unigram reference) because a 4-dimensional output table cannot express a real unigram model — the remaining gap to unigram is output calibration, not mechanism, and the gap to the trained 1k model (2.23) is the lexicon, which is what training adds. The build's own failure modes were as informative as the result: oscillatory slow modes destroy accumulation (hence zero ω on the integrators), sparse embeddings let the SiLU gate choke the wave path (a dense-bias fix), and a LayerNorm mean-shift can counterfeit a case channel — each a named mechanism, found with the same decomposition tools used on the trained model. Form by hand, content by data.

7 Where the organization comes from

The structures above exceed naive task requirements; we audited whether they are required at all.

Solution-load $\neq$ task-requirement (the audit's core result, one table). Destroying the structure in the trained solution is costly; retraining without it is not. Eval-CE deltas:

contradicted by the tiny term counts: at the 10m tier the dominating terms sit at their positional base rate (boundary bytes hold 21% of the top-25 terms vs a 20% base rate); at smaller tiers boundary positions are moderately *over*-represented (28–47% vs 20–24%), consistent with the clock concentrating credit at resets rather than contradicting it. Either way the clock sets the *reach* of every term through the warped-distance kernel.

structure	surgery (this solution)	retrain (the task)	verdict
named feature tables	+5.5 (to +0.05 at 10m)	−0.002	not required
ρ timescale ladder	+0.13–0.33	−0.001	not required
mode bank (N=6 → 1)	—	+0.038, $p = 0.002$	small, real
oscillation (ω)	—	+0.037, $p = 3\times10^{-4}$	small, real
π-harmonic prior	—	−0.002	unnecessary

Surgery used 3 seeds × 3 permutations; retrains are paired per-seed deltas (7 seeds for mode bank, 5 for oscillation, 3 for the nulls, which sit inside their detectability bounds). A full mode permutation changes nothing ($+10^{-8}$ nats, exact gauge at every tier): *a mode is a coherent object* — the value is the within-mode consistency of frequency, timescale, and gains, not the ordering. The scale story runs the same way on both sides of the audit: surgically, the tables and the ladder carry the solution at small scale and are nearly free at 10m (+2.5 → +0.05 and +0.35 → +0.006 nats); on retraining, no variant exceeds 0.031 nats at any tier (10k/1m/10m, seed 42). The frequency–mode consistency is the exception that stays load-bearing everywhere (+1.6–+3.1).

Learning curves: "not required" at every budget, not just at convergence. A retrain null at full budget is compatible with the structure being an inductive-speed advantage paid for in data. The retrain logs answer this directly: evaluated on one shared eval surface (the ablation harness's own, applied to the campaign's intermediate checkpoints), the nofeat and flat-ω retrains track the baseline's eval CE at every measured budget from 25k steps on — step-matched gaps within ±0.01 nats (flat-ω opens a small early gap, $+0.012 \pm 0.008$ at 25k steps, closed by 50k), and the median data-overhead ratio is 0.84× / 1.08× — no measurable data penalty for removing the structure (`W12_learning_curve.py`; the same-surface protocol matters: the campaign and ablation logs use different eval sets, and mixing them fabricates a spurious +0.05 gap). A mismatched-prior control completes the separation: the named feature tables *present but shuffled at init* (the seven byte features permuted among themselves, trainable) retrain to baseline as well — paired per-seed deltas $+0.009, -0.006, -0.012$ nats, mean -0.003 ± 0.010 — so it is not merely that the structure can be absent; a *wrong* assignment of the same structure is equally free. And the basis itself is scaffolding: with the seven named predicates replaced by a fixed, untrainable random partition of the byte range (`scramfeat`, 2 seeds), the model retrains to baseline (−0.005 / −0.011 nats) and the *same* boundary clock emerges with the same causal profile (Δt ratios 2.2–2.7, clamp damage input-layer-concentrated) — the clock does not read the gate basis, and the gate basis carries no necessary linguistic value. The organization is the solution's, not the task's.

Origin. Claim. The frequency skeleton is an init prior (π-harmonics), not destroyed by training — and unnecessary.

Evidence. Flat-ω retrains (no harmonic prior, trainable) match the baseline loss and rebuild a frequency ladder from zero (R^2 0.83–0.96 by the geometric-fit statistic, now over the ω dimension — a different object from the timescale ladder of §3). Everything else — timescale ladder, boundary clock, gate split — is learned within a few hundred steps, without oscillation, and in analogous form across architectures. Step-0 gradient anatomy (20 batches): the embedding takes 94.7% of the initial gradient, yet the per-mode gradient of $\log\sigma$ is already non-uniform (CV 0.43 ± 0.08 / 0.28 ± 0.05) — data breaks mode symmetry immediately, prior or not.

Caveats. The initial step scale ($\Delta t_0 = 0.05$ by the softplus bias init) is a *necessary condition* for the clock: with $\Delta t_0 = 0.5$ it never emerges (10k steps). The emergence ordering, from measured trajectories (the figure illustrates; the claims rest on the numbers): timescale organization within the first 250 steps; content-residual growth next (gate named-share collapses 1.0 → 0.02–0.14); the

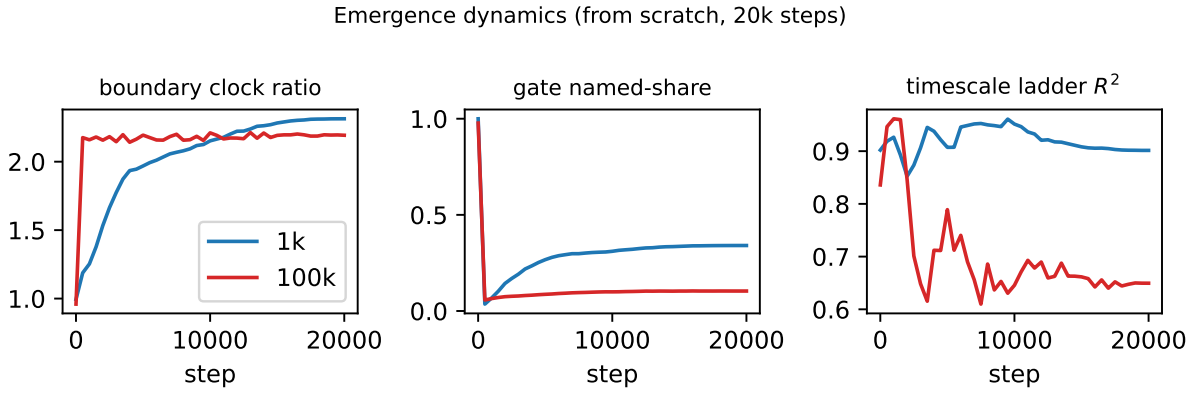


Figure 5: Emergence trajectories (from-scratch runs, 20k steps, 1k and 100k tiers): the timescale organization is immediate, the clock follows within hundreds to thousands of steps, the gate split re-equilibrates slowly. *The organization is built early, in a fixed order.* Shown for illustration of the emergence order; these are constant-learning-rate diagnostics, and the noisy 100k ladder-R^2 trace carries no claim.

boundary clock emerges by 250 steps at 100k and 2–4k at 1k; the gate split re-equilibrates slowly to a plateau (0.34 at 16–18k steps, 1k tier).

Retractions (lessons learned). Early drafts claimed the organization was task-required (retrain test: no) and that channels form antipodal specialist pairs (position-shuffle null: no). The causally load-bearing specialist singletons survive — e.g. the L0 boundary-detector channel, +1.08 CE when zeroed. A masking bug in the first-pass implicit-boundary and per-layer clamp batteries, caught in review, is also corrected here: it understated the L0 necessity clamp (+0.556 reported; $+2.57 \pm 0.48$ corrected), understated the 100k per-layer concentration ($+0.56/+0.13$ reported; $+3.19/+0.80$ corrected), and perturbed the 1m per-layer spread and the implicit-clamp comparison; all clamp numbers in the paper are the corrected ones.

Why a language model at all. A synthetic task would show that hand-set gates *can* organize; only real text shows whether the organization gradient descent builds on language statistics is the one we claim. The substrate has to touch the data whose structure it is meant to reveal — the clock, the ladder, and the split are answers to byte text, not to a toy distribution.

Statement. The organization is the path of least resistance under this parameterization — immediately and reliably found (5-seed convergence of coarse structure and of the sorted mode-value distribution, CV 3–15%) — and barely required by the task (within this parameterization family). That is precisely what makes it analyzable: the model is a readable record of an equilibrium this function family makes accessible — offered by the parameterization, not demanded by the data. A residual circularity deserves to be named: the retrain test stays *inside* this parameterization, so "path of least resistance" is established within the family — the cross-parameterization edge is left open: the Mamba-2 analogy (§3.2) is an existence observation with a granularity confound, not evidence about requirement.

8 The benchmark substrate, demonstrated

Three quantities, kept separate. *Arithmetic attribution* (what our decomposition gives): the exact sum of the model's own computed parts — no causality. *Removal sensitivity* (what LOO gives): the margin change when a byte is removed — exact for that question, by direct computation. *Causal contribution* (what interventions give): the effect of perturbing a mechanism — measured in §3.2, not in this section. The benchmark scores methods against arithmetic attribution; it says nothing about the other two.

Five standard post-hoc attributions score against the arithmetic ground truth on three decision categories (50 contexts each, plus the canonical case of §3.5):

method (Spearman ρ vs exact)	morphology (50)	word-initial (50)	post-punct (50)
gradient × input saliency	0.80 ± 0.09	0.78 ± 0.10	0.65 ± 0.11
integrated gradients	0.77 ± 0.10	0.69 ± 0.14	0.51 ± 0.12
attribution patching (EAP)	0.30 ± 0.28	0.43 ± 0.18	0.16 ± 0.21
replacement (random byte)	0.54 ± 0.19	0.50 ± 0.18	0.36 ± 0.18
leave-one-out (delete)	0.42 ± 0.18	0.27 ± 0.16	0.30 ± 0.16

The ranking is stable across all three categories: gradient methods track the arithmetic best, replacement-perturbation sits in the middle, deletion worst — and the whole grid degrades at post-punctuation decisions. Attribution patching [18] — the method family that targets causal contribution, run here as single-position corruption with the standard corrupted-run gradient — lands between the gradient rows and the perturbation rows (ρ 0.16–0.43). Under the three-way distinction above this is the expected place for it: its answer is a third quantity (an approximation of the counterfactual), and the table quantifies how far that answer sits from the arithmetic readout it is often identified with. One confound we flag rather than resolve: gradient×input and the exact arithmetic share local-linear structure (both are first-order objects of the same forward pass), so part of their agreement is expected a priori. The cells that carry information beyond that shared structure are the post-punctuation row, where every method degrades, and the removal-based rows, which diverge everywhere.

A fair objection sharpened the finding: LOO answers "removal sensitivity", and for that question it is *exact* — the margin change under removal is directly computable, there is nothing for LOO to be wrong about. The benchmark's content is therefore not "LOO is wrong" but "removal $\neq$ contribution": the two questions diverge measurably ($\rho \approx 0.3$), and a practitioner who reads removal sensitivity as decision attribution is misled by exactly that much. The gradient methods track the model's arithmetic far better than any counterfactual method, and attribution patching — the entry that explicitly targets causal contribution — lands between the two groups: a third quantity, measured here rather than asserted. A linear probe concludes "no word identity" while the substrate shows the information present but nearly linearly inaccessible (§5). The section's claim is a measurement, not a new attribution method: on a substrate where the answer is known, standard tools can be graded.

9 Related work

KANs [1] and their wavelet variants [2] motivate edges as functions; SSMs and selective scan [3, 4, 5] motivate the recurrent view — WSKAN identifies the two (an edge wavelet is an SSM impulse response). Mechanistic interpretability [6] and causal mediation [7] supply the post-hoc methods

we benchmark against; the documented unreliability of probes and attention-based attribution [9, 10] is what motivates having a ground truth at all. Within SSM interpretability specifically, the closest measurement is the Mamba-2 state sink of [14]: boundary-token Δt-gate specialization at 2.7B scale, with the caution that perturbing the probed Δt moves loss by at most 0.16 nats while suppressing the same heads' output moves it by 1.5–7.3 — an independent, much larger instance of the removal/probe dissociation our benchmark measures (§8). Their boundary class (BOS/newline, head-pooled Δt) differs from ours (spaces, per-channel Δt): they find Δt *elevated* at their boundary tokens where we measure it *suppressed* at spaces in Mamba-2's deep layers (§3.2); the two are compatible — different boundary classes, granularity, and scale — and the contrast is part of the SSM-family variation that section reports. Circuit discovery transfers to Mamba: [15] partially reverse-engineers an IOI circuit and finds entity names stored *linearly* in the SSM state at 370m scale — against our linear-inaccessibility finding at byte scale (§5); the difference (entity names in a templated task at 370m vs word identity in running text at $\leq$10m) is itself a scale-and-task statement consistent with both. The hidden-attention view of selective scan [16] and the recall scaling laws of [17] bound what SSM state can store, adjacent to our scale-level no-recall observation. Gradient-based attribution patching [18] is the method family §8 grades against exact answers. Closest in spirit is the edge-function audit of a B-spline KAN language model [11], which reconstructs, ranks, and prunes all feed-forward edge functions of a small LM and finds them mostly active and smoothly low-dimensional (their top-4 functional PCA components explain 99.9% of edge-curve variance at their grid-2 setting — a property their grid sweep shows is basis-imposed, not learned: a fresh grid-2 init already gives 99.9%, their §4.1 control analysis). Their own caveat — that causal circuit claims would require separate interventions — is the gap our intervention battery fills. Our edges are a different object: temporal SSM impulse responses with recurrence, whose readout-gain matrices are full-rank (effective rank $\approx$ 35–37 of 40 per mode in the canonical layer) — not low-dimensional in their sense. (Full-rank gains do not make the model less readable here: the readable unit is the named mode — its timescale and frequency — and the per-mode attribution of §3.5, not the gain matrix's compressibility.) The two works are complementary views of the KAN interpretability premise: static edge audit vs recurrent mechanism anatomy. Basin selection connects to the lottery-ticket literature [8], and the post-hoc reliability debate to [12, 13]. Within this intersection — a trained byte-level LM with SSM-parameterized recurrent edge functions — we know of no prior artifact whose edge functions are individually read, causally tested, and decomposed to the logit margin; broader novelty claims we do not make.

10 Limitations

Scale and task. Byte-level, up to $\approx$10m parameters, three datasets; the anatomy is consistent across measured tiers within this KAN family, and nothing is claimed beyond them. The computation is form-only — reference-level content was not found at any tier measured, and larger scales could move it into the local window; the boundary is our operational, scale-conditional choice. **Seeds and power.** 3–7 seeds per retrain family (the headline causal battery is n=5 seeds); nulls are bounded by detectability (0.04–0.10 nats, 3–8% of CE), not exact zeros; the benchmark demo covers 3 categories $\times$ 50 contexts and 5 methods. **Coverage.** Circuit-level analyses and the surgical/retrain audit are 100k-centric (the per-tier batteries cover clock, memory, margins, truncation — not the full circuit detail). **Family.** "Barely task-required" is established within this parameterization family; the cross-family edge rests on the Mamba-2 and transformer comparisons. The trade-off is one line: at the 100k and 10m tiers the architecture costs 0.02–0.12 nats of CE against matched baselines (more at the smallest tiers), and what it buys is end-to-end auditability — that

is the whole exchange, stated without inflation. **Statistics strategy.** Effect sizes with seed-std for campaign tables; paired per-seed deltas with t-tests for retrains, Holm–Bonferroni over the 5-variant family (corrected $p = 0.0015$, 0.008); sign tests where direction is the claim; explicit null distributions where a structure claim could be a metric artifact (the antipodal retraction used a position-shuffle null). For the null retrains we report detectability instead of zero: at 80% power the nofeat / ρ-ladder / flat-ω tests would have resolved effects of ≥ 0.04 / 0.09 / 0.05 nats — "not measurable at this noise level" means smaller than that, not exactly zero. No conclusion depends on a lone borderline test.

11 Conclusion

A language model whose edges are wavelets can be read like an instrument panel, and — at small scale — its next token can be derived by transparent simulation (complete below 10k parameters, truncated above), or rebuilt by hand from the anatomy alone (§6.1). The instrument panel is real, and it is not a dictionary: the object-level anatomy transfers across decisions; the per-decision semantic inventory does not (20/20 behavioral fidelity, 2/20 dictionary transfer). The organization that makes this possible is neither required nor performance-optimal, and it is not the data's fingerprint: present or mislabeled, the task retrains without it. It is what the parameterization offers. We offer the artifact, the exact methods, a grading of five standard post-hoc attribution methods against exact arithmetic on it, and, as lessons learned, several of our own claims corrected or retracted along the way (listed in §7).

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A Units and the representative-mode tuple

All dynamical quantities live in warped-time units ($\sum \Delta t$). Marginal medians on the canonical checkpoint: L0 $\tilde{\sigma} \approx 5.7$, $\tilde{\omega} \approx 9.9$; L1 $\tilde{\sigma} \approx 5.6$, $\tilde{\omega} \approx 11.4$; $\bar{\Delta}t \approx 0.11$–$0.15$. Per-mode identities are exact by construction; medians of the correlated quantities do not satisfy them (median of a ratio $\neq$ ratio of medians). All ratios quoted elsewhere in the paper are ratios of *means* (Δt at spaces vs letters), which are well-defined point estimates. Representative exact tuple (L1, channel 0, mode 3 — an arbitrary choice; the identity holds for every mode by construction): $\tilde{\sigma} = 6.107$, $\tilde{\omega} = 13.169$, $\bar{\Delta}t = 0.153 \Rightarrow Q = 1.078$, $t_{1/2} = 0.740$ tokens, 0.321 cycles/token.

B The anatomy at every scale

tier	Δt ratio, space/letter (L1)	half-life med	half-life max	word-initial vs control	`frien`→`` `d' ``
1k	1.63 ± 0.40	0.53	4	+1.70 vs +0.00	0/5
10k	2.41 ± 0.24	0.63	2*	+4.01 vs +0.01	2/5
100k	1.95 ± 0.27	0.92	69	+2.50 vs +0.13	5/5
1m	1.71 ± 0.05	1.28	92	+1.12 vs +0.20	5/5
10m	2.48 ± 0.14	2.45	3190	+3.16 vs +0.35	5/5

(*the 10k half-life max = 2 is a single-layer-tier artifact — no second layer to host slow modes; the 10k median stays on the monotone trend regardless. 5 seeds each; UltraChat.)

The rest of the anatomy, per tier (seed 42): the clock lives on every layer at every tier (at the 6-layer 1m tier the per-layer ratios are [2.18, 1.86, 1.52, 1.56, 1.61, 1.75]); the timescale ladder is geometric at every layer (fit R^2 0.60–0.97); the byte manifold separates classes everywhere (within/between cosine gap 0.09–0.17); the gate named-share declines with capacity (0.40/0.26 at

1k/10k → 0.07/0.007 at 10m — bigger models lean less on the named tables, consistent with the retrain finding that they are not required). *The objects are present at every tier we measured; their dimensions and reliance patterns are not.*

C Performance context (byte-level cross-entropy, 5 seeds)

Context, not a claim of superiority: Mamba-2 leads 12 of the 15 cells (the four smallest tiers on all three datasets, plus 10m TinyStories and 10m UltraChat), the transformer leads 1m TinyStories and 10m WikiText, and WSKAN leads 1m UltraChat clearly and 1m WikiText by 0.0002 nats — a tie within seed noise. (An earlier draft of this table claimed the full 1m tier; the campaign numbers do not support that, and the rows are now included below.)

tier (dataset)	WSKAN	WSKAN-real	Mamba-2	transformer
100k (TinyStories)	0.776±.008	0.812±.005	**0.747±.004**	0.836±.020
100k (UltraChat)	1.184±.021	1.230±.018	**1.158±.019**	1.288±.027
100k (WikiText)	1.303±.012	1.344±.008	**1.282±.010**	1.397±.021
1m (TinyStories)	0.581±.010	0.592±.007	0.639±.008	**0.559±.012**
1m (UltraChat)	**0.934±.010**	0.957±.011	1.010±.022	0.973±.056
1m (WikiText)	**1.110±.007**	1.128±.009	1.170±.012	1.110±.029
10m (TinyStories)	0.546±.005	0.558±.004	**0.478±.007**	0.487±.009
10m (UltraChat)	0.833±.006	0.861±.007	**0.716±.007**	0.799±.027
10m (WikiText)	1.023±.017	1.045±.019	0.948±.020	**0.940±.015**

D Hand-simulation truncation curve

50 held-out 24-byte contexts per tier; terms ranked by contribution to the predicted byte's logit; "static alone" = embedding + base + norm parts only.

tier	(d, L)	history terms	static alone	median terms	≤ 25 terms	p90
1k	(4, 1)	576	10%	12	76%	38
10k	(12, 1)	1,728	14%	26	50%	135
100k	(40, 2)	11,520	26%	6	74%	97
1m	(80, 6)	17.7M	50%	1	80%	44
10m	(512, 2)	37.7M	30%	2	78%	61

The count is non-monotone in size (12, 26, 6) because it tracks decision *sharpness*, not term count: the mean top-1–top-2 margin on these contexts grows 0.94 → 1.15 → 2.37 across the three tiers, and a larger margin means fewer terms suffice to fix the argmax. The 10k tier — still single-layer but wider — sits in the least sharp regime relative to its term budget. Note the direction this implies for hand-verifiability: the complete term-by-term verification lives in the *least* sharp regime (1k, where a median 12 terms still suffice); sharpness is what makes the big tiers' decisions cheap to follow but also is what training adds.

E Training protocol

All runs are byte-level, tied embeddings, AdamW, grad-clip 1.0, cosine schedule. Datasets: TinyStories, UltraChat-200k (byte stream), WikiText-103. Campaign: ≈3-epoch budgets at the 100k

tier (2.6–3.0 epochs); capacity-tiered step caps at the other tiers (0.3–2.0 epochs), matched across models per (tier, dataset); seeds 42/7/123/2024/31337. Model tiers (wskan11: d, layers): 1k (4, 1), 10k (12, 1), 100k (40, 2), 1m (80, 6), 10m (512, 2); matched-size Mamba-2 / transformer / conv / lstm baselines. Batch 64, block 256 (TinyStories/WikiText) or 512 (UltraChat); lr 3×10^{-3} ($\leq$100k) or 10^{-3} (1m+); bf16 scan in training, fp32 in analysis. The canonical analysis checkpoint is `wskan11_ultrachat_100k_3ep_ s42`. Analysis scripts and measured numbers: `experiments/` and `experiments/figures/`.

F Reproduction map

Code and checkpoints: `https://github.com/llaa33219/wskan`. The 15 campaign checkpoints of the released family (5 sizes $\times$ 3 datasets, seed 42) are also published as safetensors on Hugging Face, `llaa33219/wskan-{size}-{dataset}` with `size` $\in$ {1k, 10k, 100k, 1m, 10m} and `dataset` $\in$ {tinystories, ultrachat, wikitext} (e.g. `https://huggingface.co/llaa33219/wskan-10m-ultrachat`); each model card carries its `config.yaml`, real generation outputs, and a runnable one-command inference path (`experiments/RELEASE_infer.py`), and the published weights are verified to regenerate those outputs byte-exactly (`experiments/RELEASE_verify.py`). Every quantitative claim maps to a script in `experiments/` (script names carry the project's version-era prefixes, V1…W12; each maps to the analysis it runs, below). Figures are generated from the checkpoints by `experiments/W11_paper_figures.py` and must be present in `paper/figures/` for compilation. Checkpoints live under `checkpoints/` (campaign tier `wskan11_*_3ep_s*`); measured numbers under `experiments/figures/`. The table below maps claim to script.

claim	script
clock field, gate splits, units (§3)	`W7BC_canonical_probe.py`
causal battery, 5 seeds (§3.2)	`W11_all_sizes_analysis.py`
clock sufficiency (false ticks; V6-era, 3 seeds)	`V6_causal_clock_full.py`
production trace	`W7BC_generation_analysis.py`
ablation + performance tables	`CAMPAIGN_aggregate.py`
all-sizes anatomy (Appendix B)	`W11_all_sizes_analysis.py`
hand-simulation (§6, Appendix D)	`W11_hand_simulation.py`
exclusion battery + probes (§5)	`W11_round5_probes.py`
interventions, init-vs-trained, dynamics (§7)	`W11_excess_structure.py`
per-tier surgery; per-layer causal clamp	`W11_surgery_all_tiers.py`
retrain test (§7)	`W11_ablation_retrain.py`
seed convergence; antipodal permutation null	`W11_round7_probes.py`
basin origin	`W11_basin_origin.py`
benchmark demo (§8)	`W11_benchmark_demo.py`
attribution-patching benchmark row (§8)	`W12_benchmark_ap.py`
Mamba-2 causal battery (§3.2)	`W11_mamba2_causal.py`
transformer boundary causal test (§3.2)	`W11_tf_causal.py`
5-seed clamp; implicit-boundary clock test	`W11_implicit_boundary.py`
5-seed implicit-boundary extension (§3.2)	`W12_implicit_boundary_5seed.py`
null-calibrated probes; readout alignment (§5)	`W12_probe_nulls.py`
learning-curve retrain analysis (§7)	`W12_learning_curve.py`
mismatched-prior retrain control (§7)	`W11_ablation_retrain.py --variant shuffeat`
depth-confound control (§3 scale section)	`W11_ablation_retrain.py --variant depth6`; `W12_depth6_analysis.p`
loss-matched Mamba-2 battery (§3.2)	`W12_mamba2_causal_lossmatched.py`
architecture-level basis scramble (§7)	`W11_ablation_retrain.py --variant scramfeat`; `W12_variant_clocks`
2-layer skeleton at 1m capacity (§3)	`W11_ablation_retrain.py --variant wide2`; `W12_variant_clocks.py`
recall-probe positive control (§5)	`W12_recall_positive_control.py`
probe robustness at 16k positions (§5)	`W12_probe_word16k.py`